

Key terms 11-15: study first

Read the textbook excerpts. Then turn this page over and answer on paper.

11. Binary | Textbook p.131

"When we want to show binary data we write it as a "binary number": that is, a number made of 1s and 0s. We can also call this base 2."

Remember: Computers represent data and instructions using binary. A written 0 or 1 represents one of two states.

12. Denary (decimal) | Textbook p.134

"We call this numbering system "base ten" because decimal has ten digits, and each position is ten times greater than the next."

Remember: Denary and decimal are two names for the same number system. Its digits are 0 to 9.

13. Hexadecimal | Textbook p.140

"Hexadecimal is a number system that uses 16 different digits, which is why we can also call it base 16."

Remember: The digits are 0 to 9 and A to F. A represents 10; F represents 15. Each hexadecimal digit corresponds to four binary bits.

14. Bit | Textbook p.132

"Binary numbers are made up of the digits 1 and 0. We abbreviate the term "binary digit" to "bit". Therefore, in computer science, one bit is a 0 or a 1."

Remember: A bit is the fundamental unit of information. The symbol b represents bit; do not confuse it with B.

15. Byte | Textbook p.132

"Inside the computer, on/off bits are organised into groups of eight. A group of eight bits is called a byte."

Remember: The symbol B represents byte. Divide a number of bits by 8 to obtain bytes; multiply bytes by 8 to obtain bits.

Textbook: OUP, Oxford International AQA Examinations: International GCSE Computer Science (2017), pp.131, 132, 134 and 140.

Check against the syllabus: binary = base 2; decimal = base 10; hexadecimal = base 16; 1 byte = 8 bits. For g210 calculations, 1 kB = 1000 bytes.

Key terms 11-15: recall

DO NOW | 10 marks | Write your answers in the spaces below.

Name: _____ Class: _____ Date: _____

Turn the study page face down and close the reading on your screen. Answer from memory. When I give you the password, check your answers and add corrections in a different colour.

11. What is the binary number system? State its base and the two digits it uses. [2 marks]

12. What is the denary number system? State its base and the range of digits it uses. [2 marks]

13. What is the hexadecimal number system? State its base and all the digits or symbols it uses. [2 marks]

14. What is a bit? State its possible values and the symbol used for bit. [2 marks]

15. What is a byte? State how many bits it contains and the symbol used for byte. [2 marks]

After checking: circle one term to revisit. Explain it to your partner without looking at the study page.